THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET C

Chapters 13 & 14: **Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Think of a number n. Add 4, multiply by 5, then subtract 20. The result is: 1

(a) $5n$ **(b)** $4n$ **(c)** $5n + 4$ **(d)** $n + 4$

2. The two-digit number with tens digit 5 and ones digit 2 can be written as: 1

(a) $10 \times 2 + 5$ **(b)** $10 \times 5 + 2$ **(c)** $5 + 2$ **(d)** 5×2

3. The sum of 47 and its reverse 74 is: 1

(a) 99 **(b)** 121 **(c)** 132 **(d)** 11

4. In a number pyramid, each block is the sum of the two below it. The bottom row is 8, 1, 9. The top block is: 1

(a) 21 **(b)** 18 **(c)** 20 **(d)** 19

5. The sum of any two consecutive integers n and $n + 1$ is: 1

(a) prime **(b)** a multiple of 4 **(c)** even **(d)** odd

6. The area of a rectangle of length 18 cm and breadth 11 cm is: 1

(a) 198 cm² **(b)** 209 cm² **(c)** 29 cm² **(d)** 58 cm²

7. The area of a triangle with base 16 cm and height 15 cm is: 1

(a) 136 cm² **(b)** 31 cm² **(c)** 120 cm² **(d)** 240 cm²

8. The area of a parallelogram with base 9 cm and height 7 cm is: 1

(a) 31 cm² **(b)** 70 cm² **(c)** 32 cm² **(d)** 63 cm²

9. The area of a trapezium with parallel sides 15 cm and 25 cm and height 8 cm is: 1

(a) 160 cm² **(b)** 175 cm² **(c)** 320 cm² **(d)** 375 cm²

10. The area of a rhombus with diagonals 24 cm and 10 cm is: 1

(a) 34 cm² **(b)** 120 cm² **(c)** 60 cm² **(d)** 240 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Fill the pyramid: bottom row 2, 3, 5; find the top. 2

12. Find the largest product of two whole numbers with sum 25. 2

13. Two parallelograms on the same base have heights 5 cm and 8 cm. Find the ratio of their areas. 2

14. Find the side of a square whose area is 225 m². 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Show that the trick 'Think of n, add 3, square, subtract n^2, subtract 9, divide by n' always gives 6. 3

16. The area of a rhombus is 168 cm² and one diagonal is 24 cm. Find the other diagonal and the side. 3

17. Prove that the sum of any three consecutive even numbers is a multiple of 6. 3

OR

A number pyramid has the top 25 and the bottom row a, 6, c with $a = 7$. Find c. 3

18. A rectangular carpet 6 m × 4 m is laid in a room 8 m × 6 m. Find the uncovered area. 3

OR

A triangle and a parallelogram are on the same base 10 cm and between the same parallels 6 cm apart. Compare their areas. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Show that the sum of two odd numbers is even and of an odd and an even is odd algebraically. 5
(ii) Give an example.

OR

(i) Expand and simplify $(n+4)^2 - (n-4)^2$. 5
(ii) Evaluate for $n = 9$.

20. (i) Find the area of an L-shaped region made of rectangles 10×4 and 6×3. 5
(ii) Suggest another method.

OR

(i) A square and a rectangle have equal perimeter 48 cm; the rectangle's length is 16 cm. Compare their areas. 5
(ii) What do you conclude?

END OF QUESTION PAPER Total: 40 Marks

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